PORTFOLIO PROJECT · WINDOWS SECURITY

AEGISWFP

Adaptive Endpoint Firewall & Network Intelligence

See. Understand. Defend.



Juan Paolo · C# / .NET 8 · WPF · Windows Service · SQLite · 26 phases + 8 updates · v1.8

THE PROBLEM

Two people, one PC, and almost no visibility

The curious user

Windows Firewall says “on”. But which programs are talking to the internet right now, and should they be?

The parent

Parental controls stop at one device. The kids’ phones, tablets and consoles share the same Wi-Fi.

AEGISWFP answers both - from one app, and honestly.

WHO IT’S FOR

Made for parents who aren’t “tech people”

The worry

Kids hooked on mobile games like Mobile Legends, Roblox and Free Fire - and phones still on at bedtime.

The gap

Modem settings are confusing, and most phone apps only control one device at a time.

What AEGISWFP gives

One app on the family PC: plain words, simple switches, ready-made schedules - and it reaches every phone on the home Wi-Fi.

Honest tip in the app: pair it with Google Family Link or Apple Screen Time. AEGISWFP covers the home Wi-Fi; those cover mobile data.

THE PRODUCT

One app, two experiences

Security Center

Dark, dense, professional. Live connections with risk scores, firewall rules and their problems, what's exposed to the network, history, one-click response with undo.

For: security-minded users, IT

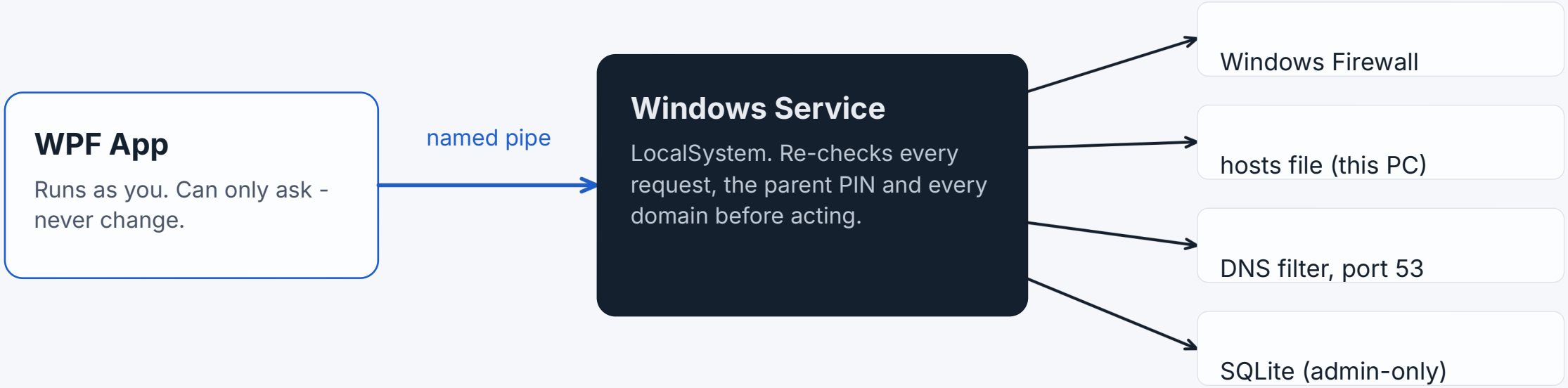
Family Shield

Light, friendly, PIN-protected. Block sites and games on this PC and every device on the Wi-Fi, with profiles and bedtime schedules.

For: parents, non-technical users

ARCHITECTURE

Only one process can change anything



Every firewall change takes one path: **Preview → Confirm → Apply → Change log → Rollback**. That goes for buttons, the terminal and Family Shield's own rules.

SECURITY CENTER · 8 FUNCTIONS, ONE PER SLIDE NEXT

See it, understand it, act on it

Live connections

Every program online, with a rule-based risk score and its reasons.

Rule intelligence

Duplicate, conflicting and too-broad firewall rules, explained.

Attack surface

What's listening, and whether the firewall really lets it be reached.

Time machine

Search past connections and spot new destinations.

Incident response

Block app, IP, port or listener - previewed, logged, undoable.

Terminal + AI analyst

Typed commands, and an optional read-only AI explanation.

SECURITY CENTER · FUNCTION 1 OF 8

Overview dashboard

What it does

The first page you see. Six status tiles, a “Needs your attention” list, and the five riskiest connections right now.

Why it matters

You know in five seconds whether anything needs you - without reading 600 firewall rules.

How it works

- Tiles: firewall, connections, suspicious processes, listening ports, rule findings, baseline
- Click any tile or “Open” to jump to its page in the sidebar
- Attention items come only from real data - when nothing is wrong, it says so

Live connections and risk scores

What it does

Lists every program talking to the network, who signed it, and a 0-100 risk score with the reasons behind it.

Why it matters

Malware has to talk to someone. This is where you catch it - and see why it looks odd.

How it works

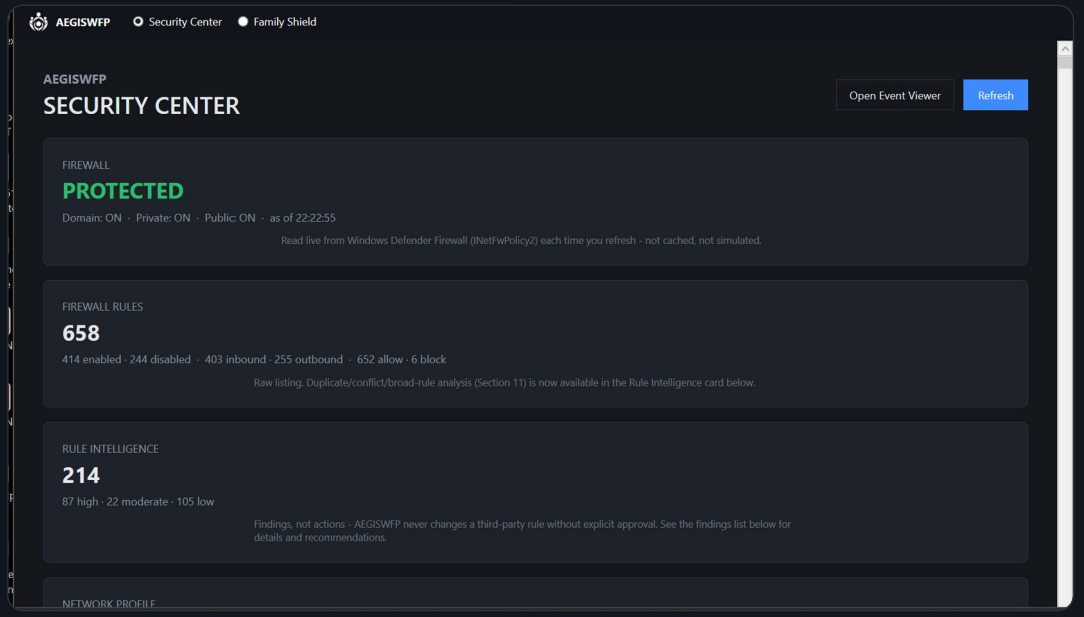
- Reads Windows' TCP/UDP tables and checks each program's digital signature
- Score = visible rules: unsigned, first time seen, unusual port, connects seconds after starting
- A behavioral baseline learns what's normal - "unusual" never means "malicious"

SECURITY CENTER · FUNCTION 3 OF 8

Firewall and Rule Intelligence

Shows if Windows Firewall is on for every network, lists all rules, and flags rules that repeat, contradict each other or allow too much.

- Create, turn on/off or delete rules
- Every change: preview → Yes → change log → Rollback
- Findings only - it never edits another program's rule by itself



Attack surface

What it does

Finds every program waiting for incoming connections and tells you whether other devices can actually reach it right now.

Why it matters

An open door you don’t know about is the easiest way in - especially on café Wi-Fi.

How it works

- “Reachable” = an enabled Allow rule covers the port, the current network profile and the program
- Remote Desktop and file sharing get their own plain-language status
- Block Listener closes a door in one click - with preview and undo

Network Time Machine

What it does

Searches past connections: which program talked to which address, and when. First-time destinations get a NEW DESTINATION badge.

Why it matters

Answers “what happened yesterday?” after something already felt wrong.

How it works

- Stored in SQLite, kept 30 days (configurable), purged daily
- Each app + destination is recorded at most once per 5 minutes - small database
- Last hour, 24 hours or 7 days, searchable by program or address

Incident response

What it does

Investigate a suspicious program, then block it, its destination IP or port in one click - and export the evidence as a file.

Why it matters

Seeing a problem is only useful if you can act on it safely.

How it works

- The Service writes the preview; you must click Yes
- Every block lands in the change log with who did it
- Rollback restores exactly what was captured before the change

SECURITY CENTER · FUNCTION 7 OF 8

AEGIS terminal

A Kali-style command line for people who prefer typing. Same powers as the buttons - no more.

- 15 fixed commands; anything else is “command not found”
- Arguments are checked: real IPs, ports 1-65535, full .exe paths
- No shell behind it - blocks still show a preview and ask first

```
┌──(aegis㉿home-pc)-[~]
└─$ connections chrome
12 connections
└─$ block-ip 203.0.113.5
[+] Block IP: applied.
└─$ rollback 3f2a9c1d
rollback requested
```

SECURITY CENTER · FUNCTION 8 OF 8

AI analyst (optional)

What it does

Asks Claude to explain what AEGISWFP found in plain words, most important first, and which button you might press.

Why it matters

Turns a wall of findings into “here’s what to look at first”.

Built to be safe

- Off by default; uses your own API key, encrypted for your Windows account
- Can only explain: its code can’t reach the firewall or the Service
- IPs and your user name hidden; “Show what will be sent” first

THE PRINCIPLE

Show what’s real - never just what a switch says.

“Blocked” lists come from reading the hosts file back, not from the settings.

“Protecting 3 devices” appears only after 3 devices really used the filter.

Categories with no trustworthy list say “not blocking yet”.

FAMILY SHIELD · FUNCTION 1 OF 10

Parent PIN and Quick Block

What it does

A parent sets a PIN, turns Family Shield on, and switches on the websites to block - YouTube, TikTok, or any site they add.

Why it matters

Only the parent can change it, and “What’s blocked right now” shows what’s really blocked.

How it works

- PIN stored only as a salted hash; 5 wrong tries lock it for a while
- Blocks go into a marked section of the Windows hosts file
- AEGISWFP reads the file back to confirm - never trusts the switch alone

Games and website categories

Games: one switch each

Roblox, Minecraft, Mobile Legends, Fortnite, Valorant, Genshin Impact, Steam - and since v1.6 the mobile favorites: Free Fire, PUBG Mobile, Call of Duty Mobile, Honor of Kings and Supercell games.

Categories: honest labels

Adult Content uses Cloudflare’s family filter. Since v1.5, Gambling (~222,000 sites) and Social Media also really block, using maintained HaGeZi lists that update themselves. The rest still say “not blocking yet”.

Only each game company’s own addresses are blocked - never shared services like Cloudflare or AWS, which would break other websites.

FAMILY SHIELD · FUNCTION 3 OF 10

SafeSearch, sites per child, and Pause now

SafeSearch and YouTube

Google, Bing and DuckDuckGo always give safe results, and YouTube runs in Restricted Mode - on every phone, even in a private tab.

Sites for one child

Each child has their own blocked list. Block TikTok for the 8-year-old without blocking it for the teenager.

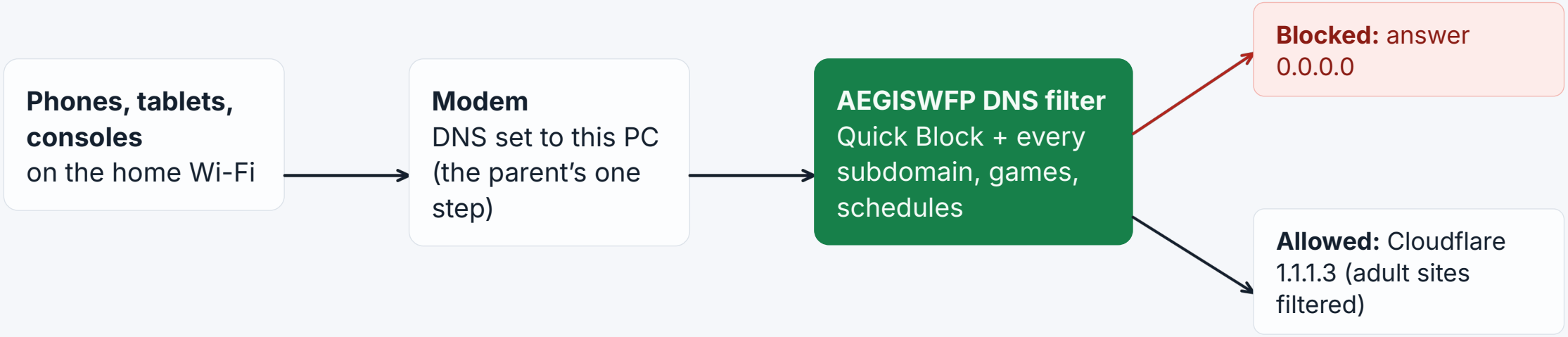
Pause now

One tap pauses a child’s internet for 30 minutes or an hour. It comes back on by itself.

How: the filter quietly answers “google.com” with Google’s own SafeSearch address, so nothing has to be installed on the phones.

FAMILY SHIELD · FUNCTION 4 OF 10

One modem setting protects every device



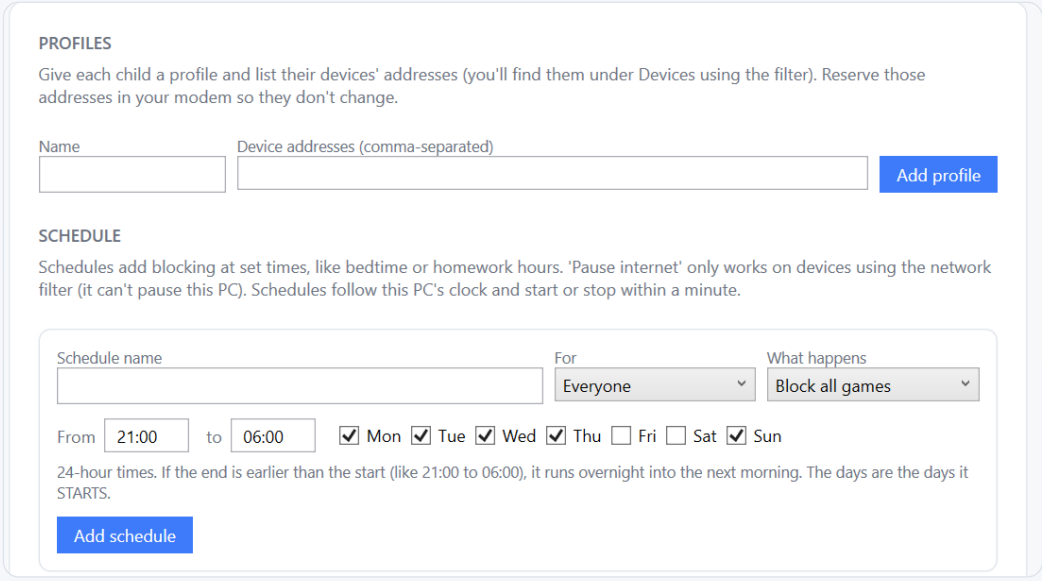
No router brand is automated - every ISP modem is different and locked down. The parent changes one setting, guided step by step, and the **Devices** page proves it worked. Since v1.5 this includes IPv6, so phones can't slip around the filter.

FAMILY SHIELD · FUNCTION 5 OF 10

Profiles and schedules

Name each child, list their devices, and set times: “Bedtime 9:00 PM - 6:00 AM, pause internet” or “Homework, block all games”.

- Devices by MAC address - AEGISWFP follows the device even when its IP changes
- Overnight times and school nights work as expected
- Never pauses the parent’s PC



FAMILY SHIELD · FUNCTION 6 OF 10

Ready-made schedules in one click

School night	Sun-Thu · 9:00 PM - 6:00 AM · pause
Bedtime	Every night · 10:00 PM - 6:00 AM · pause
Homework time	Weekdays · 4:00 - 6:00 PM · no games
Dinner time	Every day · 6:30 - 7:30 PM · pause
Weekend bedtime	Fri-Sat · 11:00 PM - 7:00 AM · pause
School mornings	Weekdays · 5:00 - 7:30 AM · no games

No typing times, no ticking days. Pick a preset, pick the child, click once.

It becomes a normal schedule, named like “School night (Mia)”, that can be switched off or deleted.

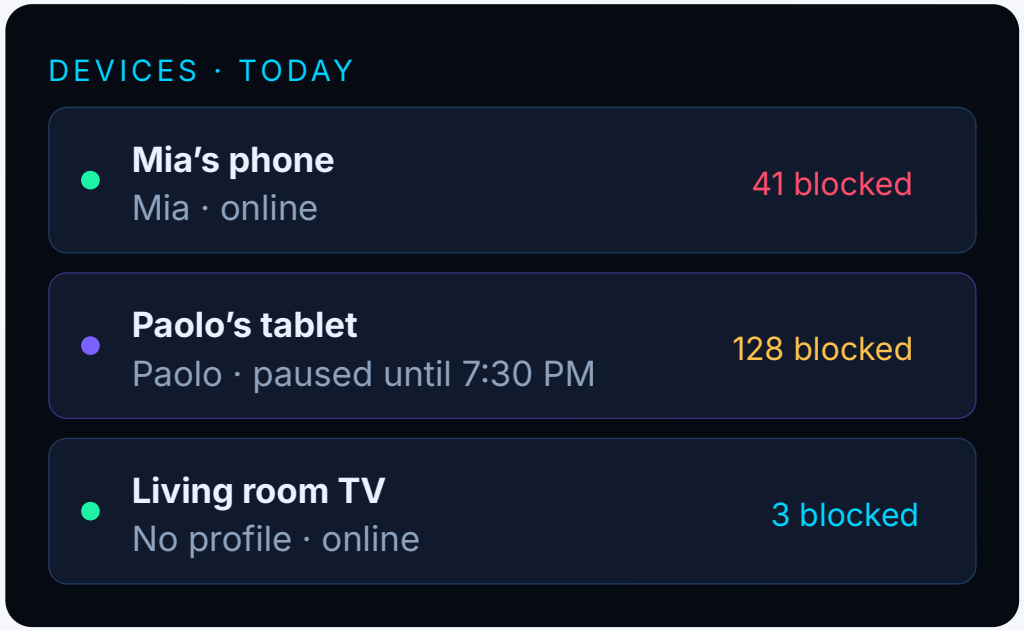
Times are shown the 12-hour way most parents read.

FAMILY SHIELD · FUNCTION 7 OF 10

Devices dashboard: who’s online, what got blocked

Every phone, tablet and TV using the filter gets a card: whose it is, whether it’s online, and how much was blocked today.

- Pause or resume a child right from the card
- Unknown devices stand out, so a parent notices a new phone
- Same device over IPv4 and IPv6 shows once



FAMILY SHIELD · FUNCTION 8 OF 10

Weekly report: is it getting better?

For each child: game attempts blocked, how long the internet was paused, and a simple trend against last week.

It counts only - it never lists which sites a child visited. Kept on the PC for 60 days, then deleted.

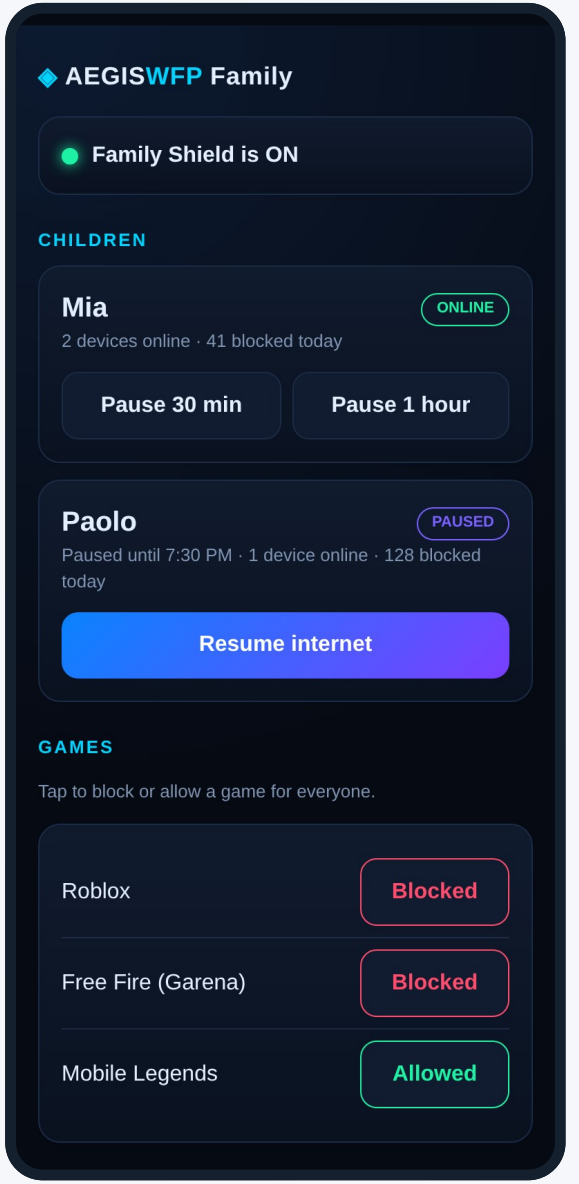


FAMILY SHIELD · FUNCTION 9 OF 10

Phone companion: control it from the couch

The parent scans a QR code on the PC with their phone camera. A page opens - nothing to install from an app store.

- See at a glance if Family Shield is on
- Pause or resume each child’s internet
- Block or allow a game for everyone
- Works only on the home Wi-Fi; up to 5 phones



AEGISWFP · 24 · Screenshot: the real phone page, test data

FAMILY SHIELD · FUNCTION 10 OF 10

A setup guide in plain words

A · Change the modem

Best: protects every device at once. Step-by-step menus for Huawei, ZTE and FiberHome modems, with a warning that PLDT modems need the admin account.

B · Change each phone

When the modem is locked: set the Wi-Fi DNS on each child’s phone. Shorter, but one device at a time.

C · Your own router

Put a parent-controlled router behind the ISP modem and set its DNS once.

The guide ends with a check: the child’s phone should appear on the Devices page. If it doesn’t, the guide says which step to look at again.

SAFETY RULES

A parental control must never break the house

01

Never cut anyone off

Turned off while devices still use it? It keeps answering, blocking nothing, until nobody has used it for 30 minutes.

02

Never lock out the parent

A “Pause internet” schedule never pauses the PC that runs AEGISWFP.

03

Never damage the hosts file

Only lines inside AEGISWFP’s own markers change. Everything else stays byte-for-byte, with a backup.

04

Never be an open resolver

The DNS filter answers home-network addresses only, rate-limited per device.

SHIPPING IT

Install, update, uninstall

Install

One Setup.exe. The Service starts with Windows and restarts itself if it ever crashes. .NET comes built in.

Update

Run the new Setup over the old one. PIN, settings and history stay.

Uninstall

Removes site blocking and filter rules, then asks about your own rules and saved data. Nothing left behind by accident.

Built with Inno Setup by one script: build-release.ps1 runs the tests, packages the app and makes the installer.

SECURITY BY DESIGN

Every threat has a mitigation and a test

Threat	Mitigation
Command injection (terminal)	Fixed verb list, typed arguments, no shell behind it
Hosts-file injection	Strict domain allow-list, checked in the Service and again before writing
Parent PIN guessing	PBKDF2 with 210,000 iterations, plus a lockout that grows to 15 minutes
Open DNS resolver	Home-network clients only, Private-profile rule, per-device rate limit
AI analyst overreach	Its project references Core only: no tools, no IPC, answer shown as text
Local data disclosure	Data folder restricted to SYSTEM and Administrators

SECURITY BY DESIGN · PHONE COMPANION

How the phone companion stays safe

STEP 1

One-time QR code

The key sits after “#” in the link, which browsers never send. The QR is hidden from screenshots, hides after 2 minutes, and expires in 10.

STEP 2

Swapped on first scan

The phone immediately trades it for a new key. A photo of the QR is useless afterwards.

STEP 3

Every tap is signed

HMAC-SHA256 with the time and a one-use random value. Copied or old requests are refused.

STEP 4

Strict server

Home network only, 4 KB requests, 60 per minute, strict page rules, last address shown to the parent.

Honest limit: it uses plain HTTP on the home Wi-Fi. Someone already on the Wi-Fi can’t steal the key or repeat a tap, but could read the status or fake the page. HTTPS is on the list.

ENGINEERING

Built in small, tested, documented steps

34

steps: 26 phases + 8 updates, each with notes, tests and limits

12

.NET projects with enforced boundaries

~25k

lines of C# (with tests), plus ~3.7k of XAML

415

xUnit test methods, including real sockets and SQLite

559

tests passing in the build check (v1.8)

16

bugs and weak spots found and fixed before release

ROADMAP

From architecture to installer - then updates

PHASES 0-4

Foundations

Threat model, App/Service split, firewall read and write with rollback

PHASES 5-9

Visibility

Connections, signatures, SQLite history, baseline, risk engine

PHASES 10-13

Intelligence

Rule analysis, attack surface, time machine, incident response

PHASES 14-18

Family Shield

PIN, websites, games, profiles and schedules, whole-network filter

PHASES 19-20

Power tools

AEGIS terminal, optional read-only AI analyst

PHASES 21-26

Ship it

Hardening, testing, installer and docs - then updates v1.1 to v1.8

AFTER VERSION 1.0

Eight updates, each one tested and shipped

v1.1
Security dashboard
One overview of the PC's health

v1.2
Profiles by MAC
Follows a phone even when its IP changes

v1.3
Neon look + Devices
Whole-app redesign and device dashboard

v1.4
SafeSearch + Pause now
Safe results, sites per child, one-tap pause

v1.5
IPv6 + real lists
No IPv6 gap; Gambling and Social Media block

v1.6
For parents
Mobile games, presets, setup guide

v1.7
Weekly report
Counts per child, trend vs last week

v1.8
Phone companion
Pause and block from the couch

WHAT MAKES IT DIFFERENT

The pieces exist. Putting them together for parents doesn’t.

One app, the whole house

A firewall view of the PC and a family filter for every phone, in one Windows app.

Honest status

It says “protected” only after checking. Otherwise it says “not yet”, and why.

Undo and preview

Every change is shown before it happens, and can be rolled back.

Private by default

Counts, not browsing history. Everything stays on the family PC.

Fair comparison: AdGuard Home filters a network and GlassWire shows a PC’s connections - both are great, but made for people comfortable with tech.

REFLECTION

What I learned, and what's next

Learned

Privilege separation is a design decision, not a feature.

Pure core logic makes Windows-heavy code testable anywhere.

Saying "not yet" honestly builds more trust than a fake green light.

Build for the real user: parents need plain words, not more features.

Next

A code-signed installer (Certum or SignPath), so Windows stops warning at setup.

HTTPS for the phone companion, and testing on real iPhones.

Real families trying it - their feedback decides what comes next.



AEGISWFP

See. Understand. Defend.

Thank you - questions?

Source, docs and threat model: the AEGISWFP repository · [demo video link]